Accepted: 13 March 2017

DOI: 10.1111/sms.12877

ORIGINAL ARTICLE

WILEY

Changes in muscle architecture of biceps femoris induced by eccentric strength training with nordic hamstring exercise

D. Alonso-Fernandez[1] | **P. Docampo-Blanco**[2] | **J. Martinez-Fernandez**[3]

[1]Department of Physical Education and Sports, Faculty of Science Education and Sport, University of Vigo, Pontevedra, Spain

[2]Real Club Celta de Vigo S.A.D., Vigo, Spain

[3]Football Club Rubin Kazan, Kazan, Tatarstan republic, Russia

Correspondence
Diego Alonso-Fernandez, Department of Physical Education and Sports, Faculty of Science Education and Sport, University of Vigo, Pontevedra, Spain.
Email: diego_alonso@uvigo.es

Eccentric strength training alters muscle architecture, but it is also an important factor for the prevention of hamstring injuries. The purpose was to determine the architectural adaptations of the biceps femoris long head (BFlh) after eccentric strength training with nordic hamstring exercise (NHE), followed by a subsequent detraining period. The participants in this intervention (n=23) completed a period of 13 weeks consisting of a first week of control and prior training, followed by 8 weeks of eccentric strength training with NHE, and concluding with a 4-week period of detraining. The architectural characteristics of the BFlh were measured at rest using two-dimensional ultrasound before (M1—week 1) and after (M2—week 9) the eccentric strength training, and at the end of the detraining period (M3—week 13). The muscle fascicle length significantly increased (t=−7.73, d=2.28, P<.001) in M2 compared to M1, as well as the muscle thickness (t=−5.23, d=1.54, P<.001), while the pennation angle presented a significant decrease (t=7.81, d=2.3, P<.001). The muscle fascicle length decreased significantly (t=6.07, d=1.79, P<.001) in M3 compared to M2, while the pennation angle showed a significant increase (t=−4.63, d=1.36, P<.001). The results provide evidence that NHE may cause alterations in the architectural conditions of the BFlh and may have practical implications for injury prevention and rehabilitation programs.

KEYWORDS

fascicle, hamstring injuries, injury prevention, muscle adaptation, nordic curl, nordic hamstring exercise, ultrasound

1 | INTRODUCTION

The architectural characteristics of a muscle largely determine its ability to produce force[1]. The pennation angle (PA), muscle thickness (MT), and fascicle length (FL) are architectural variables which vary when a muscle is subject to mechanical stimuli.[2-5] These variations affect the function and the risk of muscle injury; thus, it is important to understand their nature and potential applications in rehabilitation and prevention.[2,6]

Hamstring injuries present a great impact on sport practice,[7] counting for 12%-29% of all injuries in event categories involving high-intensity running,[8,9] in addition to reporting a high recurrence rate ranging 22%-34%.[9,10]

The scientific literature suggests two main mechanisms in hamstring lesions. The first occurs when the muscle is subject to extreme joint positions.[11] The second occurs in high-speed running.[10,12] During the terminal swing phase of the gait cycle, the hamstrings are required to eccentrically decelerate the knee extension and hip flexion before weight-bearing.[13] At this stage, the hamstrings reach their maximum length, with the biceps femoris long head (BFlh) reaching approximately 110% of its length with respect to the bipedal anatomical position.[7] This elevated eccentric stress contributes to high rates of hamstring injuries during any running movement,[9,14] mostly in the BFlh.[15,16]

Lesions in the BFlh seem to have an influence on its architecture, reducing the FL and increasing the PA, with respect

© 2017 John Wiley & Sons A/S.
Published by John Wiley & Sons Ltd

to the uninjured contralateral muscles.[17,18] These changes alter the muscle function, as a greater length of the fascicle is associated with a higher maximal velocity of contraction[1] and a lower incidence rate of injuries.[17,18] Eccentric strength training became an effective method for prevention.[19,20] Among other characteristics, it demonstrated its ability to increase muscle FL values and to reduce PA.[4,6] These architectural changes are reversible after a detraining period without eccentric mechanical stimuli.[6]

Several studies examined the changes in the muscle architecture by applying eccentric loading assessed by isokinetic dynamometry.[4,6,21,22] However, the use of isokinetic systems presents a major drawback in the everyday life of athletes, that is, their limited accessibility due to the high cost and complex handling.[23] Moreover, they are unfeasible as daily training equipment (especially for large groups of athletes), their use is time-consuming,[23] and they can also be considered "non-functional" because they bear little resemblance to actual sporting activities, being used in the sitting position.[4,6,21,22] Consequently, there is a need to evaluate alternative eccentric hamstring training systems that can become widespread in the daily practice routine of athletes.

The eccentric "nordic hamstring exercise" (NHE) has shown improvements in strength and dynamic control after 10 weeks of training,[24] a reduction in the maximum peak torque (when approaching full knee extension),[25] and a lower incidence rate of hamstring injuries in athletes and soccer players.[8,19,26] In addition, these exercises have shown a significant relationship between the "break-point angle" (angle at which the individual performing this exercise can no longer withstand the increased gravity and falls to the ground) and maximum peak torque of the hamstring measured with isokinetic dynamometry[23] and its ability to change FL in the BFlh,[27] thus suggesting its effectiveness as a basis for a functional eccentric training for athletes. NHE present a simple learning technique, which is performed with the athlete's body mass without requiring additional equipment or material and can easily be applied individually or in groups.

Due to the high incidence of the hamstring injuries and specifically in the BFlh,[15,16] it is relevant to observe the changes that occur in its architecture after an eccentric loading training. However, given the difficulties encountered in using the isokinetic equipment in the actual and everyday training routine of athletes, an interesting fact is to check whether an easily applicable eccentric exercise, such as NHE, produces architectural changes in the BFlh muscle.

2 | MATERIALS AND METHOD

2.1 | Participants

The study involved 23 male participants practicing recreational physical activity with no history of lower limb injury in the past 12 months and no previous experience in eccentric training (Table 1). All participants provided the written informed consent form before carrying out the protocol, and the previous training session took place at the Faculty of Education and Sport Sciences, University of Vigo, Spain. The ethical approval for the study was granted by the Autonomous Ethics Committee of Research in Galicia, Xunta de Galicia, General Secretariat, Ministry of Health (Spain), in agreement with the provisions of the Declaration of Helsinki (reference number: 2016/158).

2.2 | Research design

The study lasted 13 weeks. At week 1, participants undertook an initial assessment of the BFlh muscle architecture (M1) and later, two sessions of familiarization with NHE to ensure proper technical execution. Next, an 8-week period of eccentric training with NHE was performed. At week 9, the second assessment of the muscle architecture (M2) was performed. Next, the participants underwent 4 weeks of detraining, avoiding any kind of mechanical stimulus of eccentric nature. After this period, at week 13, the third assessment of the muscle architecture (M3) was carried out. All measurements were made at the same time of the day and under the same conditions for all participants.

2.3 | Assessment of the BFlh architecture

The use of ultrasound has proven to be a reliable method for assessing muscle architectural characteristics of the BFlh.[18,28] MT, PA, and the estimation of FL were determined from ultrasound images obtained along the longitudinal axis of the muscle belly using a 2D B-mode ultrasound (12 Mhz frequency, 8 cm depth; 14×47 mm field of view; GE Healthcare Vivid-i, Wauwatosa, USA). The measurement site was the halfway point between the ischial tuberosity and the posterior knee joint fold, along the line of the BFlh. Once the scanning site was determined in each participant, several anatomical landmarks were taken (ischial tuberosity, fibula head, and midpoint of the posterior knee joint fold) and photographs were taken to ensure reproducibility for future assessment sessions. All architectural measurements were performed after at least 5 minutes of inactivity, with the participant in prone decubitus position, with the hip in neutral position, and the knee was positioned passively in full extension.

To obtain the images, the ultrasound probe was placed and aligned longitudinally and perpendicular to the posterior

TABLE 1 Characteristics of participants (mean±standard error of the mean)

Group	N	Age (years old)	Weight (kg)	Height (m)
EG	23	25.2±3.3	75.5±8.1	1.76±0.09

thigh in the scanning site on the participant's skin, which was previously covered with a conductive gel. The probe was handled carefully by the sonographer (MFJ) to ensure minimal pressure and to not alter the accuracy of the measurements.[29]

After the scan, an analysis was carried out by means of image processing software, MicroDicom, version 0.7.8, developed in Bulgaria. Following the procedure conducted by Blazevich et al.,[30] six points were digitized for each of the images. MT was defined as the distance between the superficial and intermediate aponeuroses of the BFlh. PA was delimited between the intermediate aponeurosis and the direction of a muscle fascicle previously identified in the image (Figure 1). The aponeurosis angle (AA) was determined as the angle between the line marked by the aponeurosis and a horizontal line drawn along the captured image.[30,31] Finally, FL was defined as the muscle fascicle length existing between both aponeuroses. Given that the total of the abovementioned length cannot be observed in the field of view of the probe, an estimate was performed by means of an equation validated by Blazevich et al.[30] and Kellis et al.[31]

$$FL = \sin(AA + 90°) \times MT / \sin(180° - (AA + 180° - PA))$$

where FL=fascicle length, MT=muscle thickness, AA=aponeurosis angle, and PA=pennation angle.

FL was expressed in absolute terms (cm) and also relative to MT (RFL=FL/MT). All images were collected and analyzed by the same researcher (MFJ), who was blinded to the identity of the participants during the analysis. This researcher (MFJ), a member of the Spanish Society of Ultrasound in Physiotherapy (SEEFi), was previously subjected to an intra-rater study to assess the reliability of the measure. Twelve healthy subjects participated, five 2D-ultrasound images of the BFlh being performed in the same anatomical region in different sessions, obtaining reliable measurements of the different architectural variables (intraclass correlations>0.90) within the ranges recommended by previous scientific studies.[28]

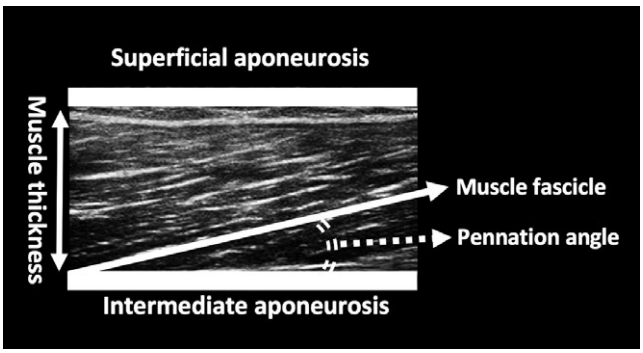

FIGURE 1 Two-dimensional ultrasound image of the biceps femoris long head (BFlh) taken along the longitudinal axis of the posterior thigh. From these images, the superficial and intermediate aponeuroses could be determined, as well as the muscle thickness and fascicle angle with respect to the aponeurosis. The estimations of the fascicle length can be performed through a trigonometric calculation, using the muscle thickness and the pennation angle

2.4 | Eccentric "nordic hamstring" exercise

NHE, also known as "nordic curl," is an exercise designed to eccentrically strengthen the hamstring muscles[12,14] and requires the assistance of a partner (Figure 2).

The athlete begins in a kneeling position with the hip and trunk elevated and fully aligned. The assistant applies pressure onto the partner's heels, ensuring their feet remain in contact with the ground during the entire execution of the exercise (Figure 2A). At the time, the athlete's trunk is lowered

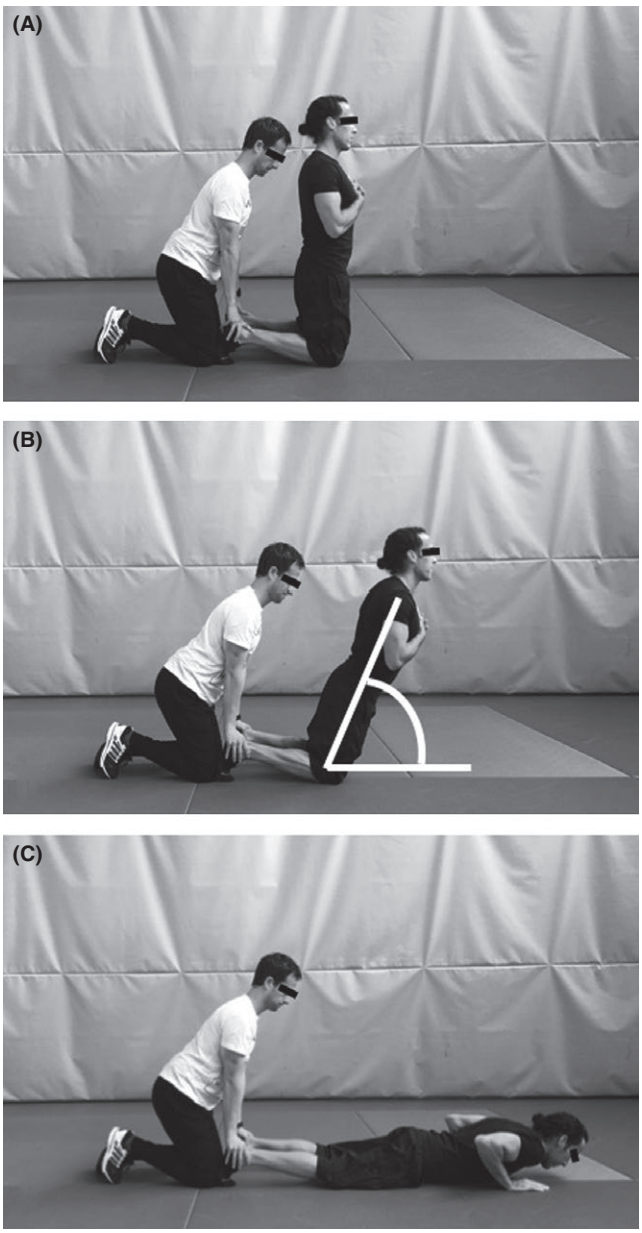

FIGURE 2 Nordic hamstring exercise: (A) start, (B) midpoint indicating the Nordic break-point angle, and (C) end

forward maintaining the initial position through a controlled extension of the knees (Figure 2B). This fall is performed at the slowest speed possible to maximize the muscle load in the eccentric phase. Hands and arms are used to buffer the fall after the break-point angle (angle at which the athlete could no longer withstand the increased gravity and falls to the ground; Figure 2C) and to return to the kneeling position minimizing the load of the concentric phase.[24] In the present study, all participants were instructed properly on the correct execution technique of NHE 2 days before training, to maximize their effectiveness. Moreover, graduates in Physical Activity and Sport Sciences supervised the training sessions to ensure the quality and consistency of the execution of each participant at all times.

2.5 | Intervention

The intervention was designed with the aim of promoting a progressive assimilation of the eccentric mechanical stimuli and reducing delayed onset muscle soreness (DOMS) in participants.

Each training session lasted between 10 and 20 minutes, depending on the training volume of sets and repetitions according to the evolution of the program (Table 2). All sessions were separated by at least 48 hours and preceded by 12-minute standardized warm-up self-loading exercises. All participants were provided with constant visual and verbal feedback, to contribute to their motivation and correct technical execution.

2.6 | Statistical analysis

All statistical analyses were performed using the SPSS software, version 24.0 (IBM Corporation, Chicago, IL). Where appropriate, data were screened for normal distribution using the Shapiro-Wilk test and homoscedasticity through Levene's test. The Greenhouse-Geisser adjustment was applied when the assumption of sphericity was violated ($P<.05$ for Mauchly's test of sphericity). One-way repeated-measures ANOVAs were used to determine training-induced changes in BFlh architecture (FL, RFL, MT and PA). The within-subject variable was time (M1, M2, and M3). In addition, post-hoc *t*-tests with Bonferroni adjustments were used to establish the pairwise comparisons between measured variables (M1-M2, M2-M3). Significance was set at $P<.05$, and where appropriate, Cohen's *d* was reported as a measure of the effect size.

3 | RESULTS

The results obtained are detailed below according to the different measured variables of muscle architecture.

3.1 | Fascicle length

The length of the muscle fascicle is affected by the eccentric training protocol with the functional NHE, $F(1.77\text{-}38.92)=41.39$, $P<.05$, $n_p^2=0.65$. Regarding the pairwise comparisons derived from the post hoc analysis, there is a significant increase in FL between M1 and M2 ($t=-7.73$, $d=2.28$, $P<.001$) and a significant decrease between M2 and M3 ($t=6.07$, $d=1.79$, $P<.001$).

3.2 | Fascicle length relative to muscle thickness

The fascicle length relative to muscle thickness is affected by the eccentric training protocol with functional NHE, $F(1.74\text{-}38.21)=18.19$, $P<.05$, $n_p^2=0.45$. As for the pairwise comparisons derived from the post hoc analysis, there is a significant increase in RFL between M1 and M2 ($t=-5.08$, $d=1.49$, $P<.001$). On the other hand, there is a significant decrease in RFL length between M2 and M3 ($t=2.76$, $d=0.81$, $P<.05$).

3.3 | Muscle thickness

Muscle thickness is affected by the eccentric training protocol with functional NHE, $F(1.73\text{-}38.08)=13.64$, $P<.05$, n_p^2

TABLE 2 Eccentric training progression with NHE

Week	No. of sessions/wk	Sets	Repetitions	Total no. of repetitions	Rest between sets
1	2	2	6	24	2 min
2	2	2	6	24	2 min
3	3	3	4	36	2 min
4	3	3	6	54	2 min
5	3	3	8	72	2 min
6	3	3	8	72	2 min
7	3	3	10	90	2 min
8	3	3	10	90	2 min

=0.38. Regarding the pairwise comparisons derived from the post hoc analysis, there is a significant increase in MT between M1 and M2 (t=−5.23, d=1.54, P<.001) and a significant decrease between M2 and M3 (t=3.64, d=1.07, P<.05).

3.4 | Pennation angle

The pennation angle is affected by the eccentric training protocol with functional NHE, F(1.92-42.31)=31.03, P<.05, n_p^2=0.58. Regarding the pairwise comparisons derived from the post hoc analysis, there is a significant decrease in PA between M1 and M2 (t=7.81, d=2.3, P<.001) and a significant increase between M2 and M3 (t=−4.63, d=1.36, P<.001).

The changes in the FL, RFL, PA, and MT during the training and detraining period are shown in Table 3.

4 | DISCUSSION

This study reported changes in the muscle architecture of the BFlh in response to an 8-week eccentric training protocol with NHE and a subsequent 4-week detraining period. The main conclusion is that eccentric training with NHE resulted in a significant increase in FL and MT of the BFlh and a significant decrease in PA. In addition, a significant reduction in FL and MT was found, as well as a significant increase in PA after detraining period.

The main contribution of these findings is in the field of practical application which, in the actual training, may lead to obtaining architectural changes by means of an exercise which is easy to reproduce and perform, such as NHE. So far, the architectural changes had been studied under laboratory conditions using expensive isokinetic equipment, beyond the reach of most athletes and coaches, and unusable in large group training. In this case, the changes in the muscle architecture have been studied using an exercise which can be applied in the actual training situations, achieving results comparable to those obtained in previous studies with isokinetic dynamometry.

The observed increase in FL of the BFlh at rest is similar to the results found in the previous literature. Potier et al.,[4] after an eccentric strength training program of 8 weeks, obtained a significant increase of 34% in FL of the BFlh. More recently, Timmins et al.[6] with an eccentric training protocol of 6 weeks have obtained similar results, with a significant increase in FL of the BFlh, of 16%. In the same vein, our study showed a significant increase of 23.9% for FL of the BFlh at rest, using an 8-week eccentric strength protocol with NHE. The differences in the duration of the intervention (6-8 weeks) and in the training modality (curl leg, isokinetic dynamometry, and NHE, respectively) can explain the differences between studies, but all authors agree on the FL increase after eccentric strength training that could be due to an increase in serial sarcomeres.[32] This situation is also reinforced by a significant increase in RFL, matching the results obtained by Timmins et al.[6] and especially those obtained by Bourne et al.[27] in a recent study, where after applying 10 weeks of training with NHE, a significant increase in FL in BFlh is observed.

PA showed a significant reduction after the training period, which coincides with the results obtained by Timmins et al.,[6] but differs from the non-significant reduction of 3.1%, obtained by Potier et al.[4] although the authors themselves pointed to this result as "counterintuitive." The changes in PA seem to be very closely related to the nature of each muscle, but it is possible that increases in PA are reliant on the extent of fiber hypertrophy that occurs and that concurrent increases in FL may counter the tendency for PA to increase.[4,33]

The present study is the only one that obtained significant changes in MT of the BFlh after an eccentric training period, considering that the previous literature [6] does not provide significant changes in the above-mentioned variable in the BFlh. However, studies on the architecture of other muscle groups, such as the vastus lateralis[21,22] and rectus femoris,[21] observe a significant increase in MT after a period of eccentric strength training. In addition, Bourne et al.,[27] using magnetic resonance imaging (MRI), observed an increase in the volume of BFlh after undergoing a 10-week training protocol with NHE.

The results also showed that the changes in the muscle architecture induced by eccentric training with NHE were

TABLE 3 Changes in muscle architecture of the biceps femoris long head before (M1) and after (M2) the intervention and after the period of detraining (M3) (mean±standard error of the mean)

Eccentric protocol with NHE (n=23)					
	M1 (Week 1)	M2 (Week 9)	M3 (Week 13)	% Change M1-M2	% Change M2-M3
FL (cm)	8.17±1.83	10.12±1.85**	8.90±1.70##	23.9	−12.1
RFL	3.91±0.39	4.53±0.55**	4.27±0.49#	15.9	−5.7
PA (º)	14.74±1.91	12.55±1.34**	13.71±1.22##	−14.8	9.2
MT (cm)	2.09±0.47	2.25±0.39**	2.10±0.45#	7.7	−6.6

FL, fascicle length; RFL, fascicle length relative to muscle thickness; PA, pennation angle; MT, muscle thickness.
**P<.001 vs M1, #P<.05 vs M2, ##P<.001 vs M2.

reversed after a 4-week detraining period. A significant decrease in FL and MT is reported, whereas PA increases. These results are consistent with those found in previous studies.[5,6]

The responses to the intervention by eccentric training with NHE and subsequent detraining may be of interest for the prevention and rehabilitation of injuries to the hamstring muscles as, as explained, shorter muscle fascicles are more prone to injury than longer fascicles.[18,34] The lesions in the BFlh modify its architecture, reducing the length of the fascicles, increasing the pennation angle, and making it more sensitive to relapse.[17,18] These reasons seem sufficient to introduce eccentric strength content in the training plan of athletes at risk of suffering hamstring injuries, and this content has proven to be an element of protection and rehabilitation of these types of injuries. In addition, the reversibility of the architectural changes when the eccentric stimulus disappears indicated that the constant exposure to such stimuli can be determining to maintain its structural effect in the long run.

So far, these potential benefits of eccentric training were hardly transposed to the daily routine of athletes, as they had been obtained with specialized equipment and conditions beyond their reach.[4,6,21,22] In our view, this can be the main contribution of this study for practical purposes: having obtained a method of eccentric training of the hamstrings with an exercise such as NHE, which can be easily applied (without additional equipment) and of short duration (between 10 and 15 minutes, 2-3 sessions per week), proving to be effective for modifying the variables of the muscle architecture of the BFlh.

The authors acknowledge that there are some limitations in the current study. First, the architecture evaluation using 2D ultrasound requires a certain degree of estimation because FL is not entirely visible in the ultrasound image. Although the estimation equation used in this study was validated,[31] there is still a potential error that should be reduced in future works based on 2D ultrasound. Secondly, only the BFlh architecture was evaluated, and considering that each hamstring has unique architectural characteristics, it would not be appropriate to generalize the results to all knee flexors.

Further research will be needed to establish the application conditions of this training type in order to achieve long-lasting architectural effects in different populations of athletes, but, in our opinion, the results may have practical implications in the prevention of BFlh injuries and in rehabilitation programs that may take into account the architectural alterations caused by training as a factor that may lower the risk of future injury.

PERSPECTIVES

The mechanical stimuli of eccentric nature seem to modify the muscle architecture of the BFlh, increasing FL. This evidence has possible impact on the prevention of hamstring injuries, as the latter decrease FL, thus increasing the risk of recurrence. However, previous results have always been obtained using isokinetic equipment, which is not readily accessible to athletes. This study is aimed at examining whether the eccentric NHE produce changes in the muscle architecture of the BFlh. The reason for choosing NHE is their simplicity and easy reproducibility; thus, they can be easily applied to any athlete individually or collectively in prevention or rehabilitation programs, without additional equipment. The adaptations produced by NHE in this study may have practical implications for injury prevention and rehabilitation programs which include the changes in muscle architecture variables.

ACKNOWLEDGEMENTS

The authors would like to thank Dr. Águeda Gutiérrez Sánchez, Dr. Margarita Pino Juste, and Dr. Antonio Rial Boubeta for their useful contributions to the production of this manuscript, as well as to the Faculty of Education and Sport Sciences, University of Vigo, for allowing the use of their facilities.

REFERENCES

1. Lieber RL, Ward SR. Skeletal muscle design to meet functional demands. *Philos Trans R Soc Lond B Biol Sci*. 2011;366:1466-1476.
2. Blazevich AJ, Cannavan D, Coleman DR, Horne S. Influence of concentric and eccentric resistance training on architectural adaptation in human quadriceps muscles. *J Appl Physiol*. 2007;103:1565-1575.
3. Narici MV, Flueck M, Koesters A, et al. Skeletal muscle remodeling in response to alpine skiing training in older individuals. *Scand J Med Sci Sports*. 2011;21(1 Suppl):23-28.
4. Potier TG, Alexander CM, Seynnes OR. Effects of eccentric strength training on biceps femoris muscle architecture and knee joint range of movement. *Eur J Appl Physiol*. 2009;105:939-944.
5. Seynnes OR, Maganaris CN, de Boer MD, di Prampero PE, Narici MV. Early structural adaptations to unloading in the human calf muscles. *Acta Physiol (Oxf)*. 2008;193:265-274.
6. Timmins RG, Ruddy JD, Presland J, et al. Architectural changes of the biceps femoris long head after concentric or eccentric training. *Med Sci Sports Exerc*. 2016;48:499-508.
7. Thelen DG, Chumanov ES, Hoerth DM, et al. Hamstring muscle kinematics during treadmill sprinting. *Med Sci Sports Exerc*. 2005;37:108-114.
8. Brooks JHM, Fuller CW, Kemp SPT, Reddin DB. Incidence, risk and prevention of hamstring muscle injuries in professional rugby union. *Am J Sports Med*. 2006;34:1297-1306.
9. Orchard J, Seward H, Orchard JJ. Results of 2 decades of injury surveillance and public release of data in the Australian Football League. *Am J Sports Med*. 2013;41:734-741.
10. Marcus C, Elliott CW, Zarins B, Powell JW, Kenyon CD. Hamstring muscle strains in professional football players: 10 year review. *Am J Sports Med*. 2011;39:843-850.

11. Askling CM, Tengvar M, Saartok T, Thorstensson A. Proximal hamstring strains of stretching type in different sports: injury situations, clinical and magnetic resonance imaging characteristics, and return to sport. *Am J Sports Med*. 2008;36:1799-1804.
12. Mendiguchia J, Alentorn-Geli E, Brughelli M. Hamstring strain injuries: are we headed in the right direction? *Br J Sports Med*. 2012;42:81-86.
13. Yu B, Queen RM, Abbey AN, Liu Y, Moorman CT, Garrett WE. Hamstring muscle kinematics and activation during overground sprinting. *J Biomech*. 2008;41:3121-3126.
14. Schache AG, Dorn TW, Blanch PD, Brown NA, Pandy MG. Mechanics of the human hamstring muscle during sprinting. *Med Sci Sports Exerc*. 2012;44:647-658.
15. Koulouris G, Connell DA, Brukner P, Schneider-Kolsky M. Magnetic resonance imaging parameters for assessing risk of recurrent hamstring injuries in elite athletes. *Am J Sports Med*. 2007;35:1500-1506.
16. Opar D, Williams MD, Timmins RG, Hickey J, Duhig SJ, Shield AJ. Eccentric hamstring strength and hamstring injury risk in Australian Footballers. *Med Sci Sports Exerc*. 2015;47:857-865.
17. Sharifnezhad A, Marzilger R, Arampatzis A. Effects of load magnitude, muscle length and velocity during eccentric chronic loading on the longitudinal growth of the vastus lateralis muscle. *J Exp Biol*. 2014;217(Pt 15):2726-2733.
18. Timmins RG, Shield AJ, Williams MD, Lorenzen C, Opar DA. Biceps femoris long head architecture: a reliability and retrospective injury study. *Med Sci Sports Exerc*. 2015;47:905-913.
19. Arnason A, Anderson TE, Holme I, Engebretsen L, Bahr R. Prevention of hamstring strains in elite soccer: an intervention study. *Scand J Med Sci Sports*. 2008;18:40-48.
20. Askling C, Karlsson J, Thorstensson A. A hamstring injury occurrence in elite soccer players after preseason training with eccentric overload. *Scand J Med Sci Sports*. 2003;13:244-250.
21. Baroni BM, Geremia JM, Rodrigues R, De Azevedo Franke R, Karamanidis K, Vaz MA. Muscle architecture adaptations to knee extensor eccentric training: rectus femoris vs. vastus lateralis. *Muscle Nerve*. 2013;48:498-506.
22. Malas FÜ, Ozçakar L, Kaymak B, et al. Effects of different strength training on muscle architecture: clinical and ultrasonographic evaluation in knee osteoarthritis. *Am J Phys Med Rehabil*. 2013;5:655-662.
23. Sconce E, Jones P, Tuner E, Comfort P, Graham-Smith P. The validity of the nordic hamstring lower for a field-based assessment of eccentric hamstring strength. *J Sport Rehabil*. 2015;24:13-20.
24. Mjølsnes R, Arnason A, Osthagen T, Raastad T, Bahr R. A 10-week randomised trial comparing eccentric vs concentric hamstring strength training in well-trained soccer players. *Scand J Med Sci Sports*. 2004;14:311-317.
25. Clark R, Bryant A, Culgan JP, Hartley B. The effects of eccentric hamstring strength training on dynamic jumping performance and isokinetic strength parameters: a pilot study on the implications for the prevention of hamstring injuries. *Phys Ther Sport*. 2005;6:67-73.
26. Van der Horst N, Smits DW, Petersen J, Goedhart EA, Backx F. The preventive effect of the nordic hamstring exercise on hamstring injuries in amateur soccer players: a randomized controlled trial. *Am J Sports Med*. 2015;43:1316-1323.
27. Bourne MN, Duhig SJ, Timmins RG, et al. Impact of the Nordic hamstring and hip extension exercises on hamstring architecture and morphology: implications for injury prevention. *Br J Sports Med*. 2016;1-9.
28. Narici MV, Binzoni T, Hiltbrand E, Fasel J, Terrier F, Cerretelli P. In vivo human gastrocnemius architecture with changing joint angle at rest during graded isometric contraction. *J Physiol*. 1996;496:287-297.
29. Klimstra M, Dowling J, Durkin JL, MacDonald M. The effect of ultrasound probe orientation on muscle architecture measurement. *J Electromyogr Kinesiol*. 2007;17:504-514.
30. Blazevich AJ, Gill ND, Zhou S. Intra- and intermuscular variation in human quadriceps femoris architecture assessed in vivo. *J Anat*. 2006;209:289-310.
31. Kellis E, Galanis N, Natsis K, Kapetanos G. Validity of architectural properties of the hamstring muscles: correlation of ultrasound findings with cadaveric dissection. *J Biomech*. 2009;42:2549-2554.
32. Lynn R, Morgan DL. Decline running produces more sarcomeres in rat vastus intermedius muscle fibers than does incline running. *J Appl Physiol*. 1994;77:1439-1444.
33. Foure A, Nordez A, Cornu C. Effects of eccentric training on mechanical properties of the plantar flexor muscle-tendon complex. *J Appl Physiol*. 2013;114:523-537.
34. Fyfe JJ, Opar DA, Williams MD, Shield AJ. The role of neuromuscular inhibition in hamstring strain injury recurrence. *J Electromyogr Kinesiol*. 2013;23:523-530.

How to cite this article: Alonso-Fernandez D, Docampo-Blanco P, Martinez-Fernandez J. Changes in muscle architecture of biceps femoris induced by eccentric strength training with nordic hamstring exercise. *Scand J Med Sci Sports*. 2017;00:1–7. https://doi.org/10.1111/sms.12877